Project Deliverable 1

MECHTRON / SFWRENG 3K04: Software Development

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Introduction and Context

INTRODUCTION

This document includes the requirements, user stories, and UML diagrams for the Hospital Monitoring System (HMS). It aims to outline what the system must do, clarifies how users will interact with it, and provides a loose depiction of how the system might operate. This document will be a valuable reference throughout subsequent design and implementation phases.

CONTEXT

The Hospital Monitoring System (HMS) is intended to act as a supervisory application that connects a network of Automated Dispensing Cabinets (ADCs) to hospital personnel. With the ultimate goal to elevate patient care, the primary purpose of the HMS is to provide users — nurses, pharmacists, technicians, clinical supervisors, and administrators — with centralized information about multiple ADCs located across the facility. The software allows these users to monitor vital ADC information, such as inventory levels, active alarms, and cabinet status.

The HMS is considered a safety-critical system because malfunctions or unintended behaviours could threaten patient safety. Examples of malfunctions include incorrect user authorization, poor circulation of critical alerts, or treating an offline device as healthy.

SCOPE

The HMS is limited to the oversight of ADCs and is separate to physical cabinet operations.

In-Scope Capabilities (not exhaustive):

- Collecting operational information from ADC units
- Providing an operator dashboard for ADC monitoring and relevant information
- Supervising inventory levels across ADC network
- Receiving, managing, and dispersing alarms
- Storing historical data and generating reports

Out-of-Scope:

- Operation of ADC hardware
- Embedded ADC software
- Network infrastructure
- External hospital information systems and clinical decision processes

Functional Requirements

ADC COMMUNICATION

FR-COM-01 The system shall maintain connection with multiple ADC units at once

ALARM MANAGEMENT

FR-ALM-01 The system shall receive alarms/alerts from all ADC units
FR-ALM-02 The system shall direct alarms/alerts to the appropriate individuals
FR-ALM-03 The system shall send alerts to hospital staff via personal devices
FR-ALM-04 The system shall allow hospital staff to adjust alarm notification settings

INVENTORY MANAGEMENT

FR-INV-01 The system shall aggregate and display real-time medication inventory levels received from all connected ADC units
FR-INV-02 The system shall flag and display low-inventory items whenever an ADC inventory count falls below a predefined minimum threshold
FR-INV-03 The system shall generate an alert and queue a restock task when a medication falls below par levels
FR-INV-04 The system shall allow authorized pharmacy personnel to configure medication par levels and inventory thresholds
FR-INV-05 The system shall track inventory replenishment events
FR-INV-06 The system shall update cabinet status upon confirmation of ADC restock

REPORTING

FR-REP-01 The system shall periodically report to senior hospital staff
FR-REP-02 The system shall allow users to download reports

DASHBOARD

FR-DSH-02 The system shall provide relevant ADC data when selected on the dashboard
FR-DSH-04 The system shall automatically update an ADC's status on the dashboard
FR-DSH-05 The system shall display unreviewed log-in attempts on the IT administrator dashboard
FR-DSH-06 The system shall display unassigned requests on the clinical supervisor dashboard
FR-DSH-07 The system shall display assigned tasks on the nurse dashboard
FR-DSH-08 The system shall display assigned tasks on the pharmacist dashboard
FR-DSH-09 The system shall display completed tasks on the clinical supervisor dashboard

ADC HEALTH MONITORING

FR-MON-01 The system shall continuously verify ADC connectivity
FR-MON-02 The system shall generate an alert if an ADC fails a health check

LOGGING

FR-LOG-01 The system shall record all log-in attempts
FR-LOG-02 The system shall provide advanced search capabilities for event logs based on date ranges, user IDs, event types, and cabinet locations
FR-LOG-03 The system shall record all medication requests, dispenses, restocks, threshold breaches and dispensing failures
FR-LOG-04 The system shall record all alarm resolutions, overrides, and administrative acknowledgments performed by authorized staff across cabinets
FR-LOG-05 The system shall record all maintenance work orders, scheduled service logs, connection status, error messages, and diagnostic routines

TASK ASSIGNMENT

FR-TSK-01 The system shall allow nurses/pharmacists to submit requests
FR-TSK-02 The system shall prompt clinical supervisors to assign tasks based on received requests
FR-TSK-03 The system shall automatically delegate routine tasks

ADMIN/IT

FR-ADM-01 The system shall enable IT administrators to create/delete login profiles
FR-ADM-02 The system shall enable IT administrators to edit login profiles
FR-ADM-03 The system shall support user log-in from hospital computers
FR-ADM-04 The system shall support user log-in from personal devices
FR-ADM-05 The system shall timeout (automatically log out) after 2 minutes of inactivity
FR-ADM-06 The system shall deny access if credentials are incorrect

USER INTERFACE

FR-UIF-01 The system shall provide a centralized operator dashboard to display aggregated ADC statuses, active alarms, and pending tasks
FR-UIF-02 The system shall allow authorized users to navigate from high-level summaries to detailed views of specific ADC inventories, patient schedules, and event logs

PATIENT INFORMATION DATABASE

FR-PID-01 The system shall store patient medication schedules, dispensing histories, and profile data
FR-PID-02 The system shall scrub personally identifiable information (PII) before transmitting to external components or personal devices
FR-PID-03 The system shall link historical dispensing logs to the corresponding patient profile
FR-PID-04 The system shall fulfill database queries according to user permissions

Non-Functional Requirements

ADC COMMUNICATION

NFR-COM-01 The system shall transmit a heartbeat ping to all ADC units every 30 seconds

ALARM MANAGEMENT

NFR-ALM-01 The system shall send critical alerts to the project manager within 5 seconds
NFR-ALM-02 The system shall escalate any non-critical alert to a critical alert should the status not be updated within 1 hour
NFR-ALM-03 The system shall send a follow-up alert at 10 am to the project manager in place of the "everything is ok" message if there is an active critical alert
NFR-ALM-04 The system shall exclude patient information from alert notifications to personal devices
NFR-ALM-05 The system shall notify IT administrators if there is incomplete user profile information
NFR-ALM-06 The system shall resend an alarm after 30 seconds in the event it is unreceived
NFR-ALM-07 The system shall forward alarms to the appropriate clinical supervisor should it fail to reach the target recipient

INVENTORY MANAGEMENT

NFR-INV-01 The system shall ensure inventory data is synchronized across ADCs
NFR-INV-02 The system shall restrict modification of inventory records to pharmacy personnel on designated hospital devices
NFR-INV-03 The system shall process incoming inventory messages from up to 50 concurrent ADC units without exceeding a 2-second processing latency
NFR-INV-04 The system shall generate a non-critical alert when a medication falls below par levels
NFR-INV-05 The system shall generate a critical alarm if a stockout affects active patient schedules

REPORTING

NFR-REP-01 The system shall send daily confirmation at 10 am to the project manager that "everything is ok" when there are no active critical alerts
NFR-REP-02 The system shall include a summary of any unresolved critical alert(s) from the past 24 hours in the 10 am check-in message to the project manager

DASHBOARD

NFR-DSH-01 The system shall process and display a "Low Inventory" alert within 5 seconds of an ADC's medication count dropping below the minimum threshold
NFR-DSH-02 The system shall reflect an ADC's loss of connection within 5 seconds of a missed 30 second heartbeat window
NFR-DSH-03 The system shall display inventory updates on the dashboard within 5 seconds of transmission

LOGGING

NFR-LOG-01 The system shall record the date, time, device, user ID, password, and outcome (access granted/denied)
NFR-LOG-02 The system shall mark failed log-in attempts for review
NFR-LOG-03 The system shall record the reason of denied access for failed log-in attempts
NFR-LOG-04 The system shall mark suspicious log-in attempts for review
NFR-LOG-05 The system shall update "for review" log-in attempts once they have been checked as "reviewed" by an IT administrator
NFR-LOG-06 The system shall ensure that all recorded event logs are tamper-evident and immutable
NFR-LOG-07 The system shall store event logs in a redundant, highly available database achieving 99.99% data durability
NFR-LOG-08 The system shall append precise timestamps to every logged event with millisecond accuracy, and display them to users with a resolution of one second
NFR-LOG-09 The system shall restrict access to detailed event logs and sensitive data to hospital-issued devices/networks while maintaining basic alerts for personal devices
NFR-LOG-10 The system shall maintain immutable historical logs of all inventory adjustments, restocking actions, and threshold breaches for a minimum of one year

TASK ASSIGNMENT

NFR-TSK-01 The system shall direct requests from nurses/pharmacists to clinical supervisor(s)
NFR-TSK-02 The system shall allow clinical supervisors to return requests to the sender for clarification
NFR-TSK-03 The system shall reference routine task list to assign routine tasks to the appropriate individuals
NFR-TSK-04 The system shall register when nurses/pharmacists mark tasks as "complete"

ADMIN/IT

NFR-ADM-01 The system shall prompt IT administrators to fill in various employee details when creating a new system login (name, employee number, department, permission level, etc)
NFR-ADM-02 The system shall authorize access on hospital computer with correct user ID, password, and employee card
NFR-ADM-03 The system shall display "Credentials invalid" if access is denied on log-in
NFR-ADM-04 The system shall authorize access on personal devices with correct user ID, password, and face ID
NFR-ADM-05 The system shall only grant access on personal devices if the device has been authorized
NFR-ADM-06 The system shall only grant access on personal devices if the device is connected to the authorized hospital network
NFR-ADM-07 The system shall allow IT administrators to establish/update a user's authorized device

USER INTERFACE

NFR-UIF-01 The system shall process and display state changes (alarms, inventory updates) within 2 seconds of receiving the data from the backend system

NFR-UIF-02 The system shall automatically obscure sensitive on-screen data and lock the session after 2 minutes of inactivity

NFR-UIF-03 The system shall prohibit users from disabling system-wide critical safety alarms to ensure adherence to safety-critical standards

NFR-UIF-04 The system shall fetch, process, and render filtered event log results within 3 seconds to maintain usability during active security investigations

PATIENT INFORMATION DATABASE

NFR-PID-01 The system shall securely encrypt all patient and medication records at rest within the database

NFR-PID-02 The system shall support concurrent queries from up to 50 active ADC units/users without exceeding a 1-second query response time

NFR-PID-03 The patient information database must maintain an immutable, tamper-proof audit trail of all record retrievals to satisfy ethical privacy of medication records

NFR-PID-04 The system must ensure that historical patient data remains available and securely protected against tampering

User Stories

ADC COMMUNICATION

US-COM-01: As an IT administrator, I want to be notified immediately if an ADC loses network connection to the monitoring system so that I can restore connection before the ADC starts to affect staff and patients.

Scenario 1: Successful Heartbeat
GIVEN ADC-01 is powered on and connected to the monitoring system
WHEN the system sends its periodic health check
THEN ADC-01 responds with its status within 2 seconds
AND the dashboard displays ADC-01 as "Healthy / Online"

Scenario 2: Connection Lost
GIVEN ADC-01 was previously displaying "Healthy / Online"
AND the monitor has received no heartbeat or status message from ADC-01 for 30 seconds
WHEN the system reevaluates ADC-01's health
THEN the dashboard updates ADC-01's status to "Offline - Connection Lost"
AND a connection alert is routed to IT staff
BUT the ADC continues to dispense locally based on its last cached permissions/data

Scenario 3: Task Acknowledgement
GIVEN a "low" inventory alert for ADC-01 is shown on the dashboard
WHEN a pharmacy technician acknowledges the alert to begin the restock
THEN alert is marked "in progress" with the technicians ID
AND it stays in the queue until the ADC-01 reports that the slot has been restocked successfully

ALARM MANAGEMENT

US-ALM-01: As the project manager, I want to stay updated with the status of the system, notified daily when everything is going well, and notified immediately in the case of critical alerts such as device communication failures, medication errors, or patient medication schedules that may pose risks.

Scenario 1: No critical alerts, everything is ok
GIVEN no active critical alerts
WHEN the time is 10:00
THEN the system sends an "everything is ok" alert to the project manager
AND includes a summary of any critical alerts resolved in the past 24 hours

Scenario 2: Minor issue occurs and is resolved by authorized user
GIVEN the system is operational
WHEN a minor issue occurs (low inventory, scheduled maintenance reminders)
AND it is resolved by a technician/clinician within the appropriate time frame for that issue
THEN the alert is marked resolved by the system and doesn't require escalation

Scenario 3: Minor issue not resolved
GIVEN a minor issue has occurred
WHEN the status of the issue has not been changed within 1 hour
THEN the system escalates this to a critical alert and notifies the project manager

Scenario 4: Critical alert
GIVEN the system is operational

WHEN a critical alert occurs (lost communication, medication errors, concerning patient data)
THEN the system immediately sends notification to the project manager

Scenario 5: Critical alert unresolved at status update
GIVEN there is an unresolved critical alert
WHEN the time is 10:00
THEN the system sends a follow-up of the ongoing alert
AND includes a summary of any critical alerts resolved in the past 24 hours

US-ALM-02: As a nurse, I want the flexibility of receiving ADC alerts through multiple avenues, such as SMS message, email, or notification from a designated app. However I'm also concerned about security/safety concerns with private information potentially being available from any device anywhere.

Scenario 1: Alert doesn't involve sensitive information
GIVEN the ADC detects that a drawer wasn't closed properly
WHEN the alert reaches the monitoring system
THEN it will be directed to all relevant staff on their personal device via their preferred method

Scenario 2: Alert involves sensitive information
GIVEN a patient is scheduled for a dose
WHEN the medication is not taken from the ADC ie the nurse has forgotten to deliver it
THEN a high-level alert, absent of names and personal details, will be shared on personal devices
AND the nurse(s) can view the alert in greater detail on a hospital-authorized device and wifi

Scenario 3: Change notification settings
GIVEN a nurse wishes to change notification settings
WHEN the nurse signs into their account on a hospital-authorized device and wifi
THEN they can toggle on/off (with min. 1 ON) for all of SMS message, email, and app notification

Scenario 4: Connection Lost
GIVEN monitoring system attempts to send alarm
WHEN connection fails
THEN retry after 30 seconds
AND escalate to clinical supervisor if second attempt fails

INVENTORY MANAGEMENT

US-INV-01 As a pharmacy technician, I want to view a real-time low-inventory alarm on the dashboard so that I can restock the medication before the unit runs out.

Scenario 1: Inventory low-stock alarm reaches the monitor
GIVEN ADC-01 is online and connected to the monitoring system
WHEN a dispense event causes [some] medication to drop below the minimum threshold
THEN the system generates a non-critical low-inventory alarm
AND the operator dashboard displays a visible "Low Inventory - ADC-01 - [some] medication" alert within 5 seconds
AND the alarm event is logged with the ADC ID, medication name, and timestamp

Scenario 2: Connection Lost during dispensing
GIVEN ADC-01 is actively dispensing
AND the monitor has received no heartbeat or status message from ADC-01 for 30 seconds
THEN the monitor flags ADC-01 as "offline" on the dashboard
AND routes a critical connection alarm to IT staff for troubleshooting this ADC

Scenario 3: Task Acknowledgement
GIVEN a visible "low inventory" alert for ADC-01 is shown on the dashboard

WHEN a pharmacy technician acknowledges the alarm to begin the restock
THEN the alarm status is marked “in progress” with the technician's ID
AND the restock task stays in the queue until ADC-01 reports that the slot has been restocked successfully

US-INV-02: As a pharmacist, I want to configure and update medication par levels (maximum and minimum thresholds for drug stock) for connected ADCs so that inventory alarms reflect current clinical demand.

Scenario 1: Pharmacist successfully updates par level
GIVEN a pharmacist is logged into the system on a hospital-issued device
WHEN the pharmacist modifies the minimum threshold for [Medication X] on ADC-02 from 10 units to 20 units
THEN the system updates the inventory threshold settings for ADC-02
AND synchronizes the updated par level with ADC-02

Scenario 2: Unauthorized user attempts to modify threshold
GIVEN a non-pharmacy user is logged into the system
WHEN the user attempts to access or modify medication par levels for any ADC
THEN the system denies access with an "Unauthorized Action" error message
AND prevents any changes to the inventory configuration

US-INV-03: As a pharmacy technician, I want the monitoring system to track replenishment events in real-time so that the dashboard automatically clears low-inventory alarms once a cabinet is refilled.

Scenario 1: Restocking confirmation received from ADC
GIVEN an active low-inventory alarm and visible dashboard alert exist for [Medication Y] on ADC-03
WHEN the pharmacy technician completes restocking at ADC-03 and the unit transmits a restocking confirmation packet
THEN the monitoring system updates the live inventory count for ADC-03 on the dashboard within 5 seconds
AND clears the active low-inventory alarm and visible alert from the queue

Scenario 2: Incomplete restock below minimum threshold
GIVEN the minimum threshold for [Medication Y] on ADC-03 is 15 units
WHEN a technician restocks ADC-03 but only adds enough units to reach 10 units
THEN the system updates the inventory count to 10 units
AND maintains the active low-inventory alarm and visible alert state on the dashboard until the threshold of 15 units is reached

LOGGING

US-LOG-01: As an IT staff member or pharmacist, I want the system to automatically record all medication transactions, inventory adjustments, restocks, and ADC maintenance work orders so that complete operational and historical records are preserved.

Scenario 1: Recording medication requests and inventory adjustments
GIVEN an ADC unit processes a medication request, dispense, restock, or threshold breach
WHEN the event telemetry is transmitted to the monitoring system
THEN the event logger records the transaction details into the system database
AND maintains immutable historical logs of the inventory adjustments for a minimum of one year

Scenario 2: Recording maintenance work orders and connection errors
GIVEN a connected ADC experiences a connection drop, hardware error, or diagnostic routine failure
WHEN the system detects the failure or processes the maintenance work order
THEN the event logger records the maintenance activity, connection status, and error messages
OR flags an unhandled error log for IT technician review

US-LOG-02 As an IT administrator, pharmacist, or clinical supervisor, I want advanced search, filtering, and export capabilities for event logs so that I can efficiently review logs by date ranges, user IDs, event types, or cabinet locations.

Scenario 1: Filtering and exporting event logs successfully
GIVEN a user accesses the event log viewing interface
WHEN the user selects specific filters for date range, event type (e.g., alarms or overrides), and cabinet location
THEN the system adjusts the event log display to show only the matching records
AND provides an option to export the filtered log data for reporting

Scenario 2: Zero results returned for restrictive filter criteria
GIVEN the event log search interface is active
WHEN a user enters a combination of filters that matches no recorded events in the database
THEN the system displays a "No matching events found" notification.

US-LOG-03: As an IT administrator, I want all event logs to feature high-precision timestamps, tamper-evident immutability, and redundant high-availability database storage so that audit trails remain secure and durable.

Scenario 1: Storing events with high-precision timestamps and durability
GIVEN an operational event or transaction is captured by the event logger
WHEN the event is written to the database
THEN the system appends a precise timestamp with millisecond accuracy
AND stores the entry in a redundant, highly available database achieving 99.99% data durability

Scenario 2: Unauthorized modification attempt on historical logs
GIVEN historical event logs are stored in the secure database
WHEN an unauthorized process or user attempts to alter, falsify, or delete recorded log entries
THEN the system blocks the modification attempt due to tamper-evident immutability constraints
AND triggers a security alert to the IT administrator

US-LOG-04: As a hospital manager or IT administrator, I want the system to restrict access to detailed event logs and sensitive data to hospital-issued devices and networks while supporting basic notifications for personal devices so that hospital security and privacy standards are maintained.

Scenario 1: Accessing detailed logs from a hospital-issued network
GIVEN an authorized user is logged in via a hospital-issued device connected to the secure hospital network
WHEN the user requests access to detailed event logs, security overrides, or credential failure records
THEN the system grants full access to the requested log details

Scenario 2: Requesting sensitive log data from a personal device
GIVEN a user is accessing the system from a personal mobile device outside the secure hospital network
WHEN the user attempts to view detailed sensitive event logs or override records
THEN the system restricts access to the detailed logs
AND limits the personal device interface to basic non-sensitive system alerts only

TASK ASSIGNMENT

US-TSK-01: As a clinical supervisor, I want to manage and assign specific requests to/from nurses and pharmacists while routine tasks get assigned automatically, to maximize flow and efficiency in the hospital.

Scenario 1: Special medication request
GIVEN a patient needs a specific medication
AND the nurse recognizes it is not available in the ADC
WHEN the nurse inputs the request into the system
THEN the request will await assignment from a clinical supervisor

Scenario 2: Manual task assignment
GIVEN a request is awaiting assignment from a clinical supervisor
WHEN the task is assigned to an available employee (pharmacist, nurse, IT worker)
THEN the task will appear on the employee's dashboard
AND they can mark it complete once it is finished

Scenario 3: Confusing details in request
GIVEN a request was sent
AND the clinical supervisor is confused about its contents
WHEN the clinical supervisor returns the request
THEN the returned request will appear on the sender's dashboard, specifying the source of confusion

Scenario 4: Routine task management
GIVEN the routine task list is inputted into the system
WHEN a task needs to be assigned
THEN the system will automatically assign it to the appropriate employee
AND the task will appear on the employee's dashboard

Scenario 5: Task Complete
GIVEN the task has been completed
WHEN the task is marked complete through the system
THEN the clinical supervisor will be notified that it has been completed
AND they can verify that it has been completed

ADMIN/IT

US-ADM-01: As an IT administrator, I need the capability to add/delete user logins as well as update account permissions to stay up to date with hospital staffing changes.

Scenario 1: Add login
GIVEN a new nurse joins the team
WHEN an IT administrator accesses the system
THEN the IT administrator can create the new account, specifying permissions and credentials

Scenario 2: Update permissions
GIVEN a nurse has been moved to a different department
WHEN an IT administrator accesses the system
THEN the IT administrator can adjust the nurse's permissions to enable access to relevant information

Scenario 3: Account information incomplete
GIVEN a new nurse joins the team
WHEN an IT administrator logs off of the system
AND the system detects a user profile with missing information (ie account setup in progress)
THEN IT administrator(s) will receive an alert to complete the login setup

US-ADM-02: As an IT administrator, I want to monitor login attempts to catch suspicious activity before it is a concern.

Scenario 1: Event logging
GIVEN the system is online and connected
WHEN a login occurs
THEN it is logged

Scenario 2: Suspicious login attempt
GIVEN the system is online and connected
WHEN a failed login is detected with a user id for a nurse who is not on shift
THEN the login attempt is marked for review

Scenario 3: View login log
GIVEN the event logs are up to date
WHEN an IT administrator accesses the system
THEN the system displays the event log filtered to only display login attempts
AND failed login attempts and logins outside normal hours are highlighted

US-ADM-03: As an IT administrator, I want users accessing the system from a **hospital computer** to login with a user ID, password, and by tapping their employee ID card, to ensure secure login.

Scenario 1: User sign in
GIVEN the system is open on a hospital computer
WHEN a nurse inputs their user ID, password, and taps with their card
THEN the system grants access

Scenario 2: Nurse forgets to log out
GIVEN a nurse is logged into the system on a hospital computer
WHEN 2 minutes have passed since the last action
THEN the system automatically times out and returns to log in screen

Scenario 3: Incorrect password
GIVEN the system is open on a hospital computer
WHEN a nurse inputs their login information with a typo in the password
THEN the system outputs “incorrect user ID or password”
AND the failed login attempt is marked for review

US-ADM-04: As an IT administrator, I want users accessing the system from a **personal device** to login with a user ID, password, and face ID. The device should also have been authorized and be connected to the hospital wifi network to ensure secure login.

Scenario 1: User sign in
GIVEN a nurse’s device is authorized and connected to hospital wifi
WHEN a nurse inputs their user ID, password, and uses face ID
THEN the system grants access

Scenario 2: Device not connected to network
GIVEN a nurse’s device is authorized, but using data instead of the hospital wifi
WHEN a nurse inputs their user ID, password, and uses face ID
THEN access is not granted
AND a message is outputted encouraging the user to establish authorized connection
AND the failed login attempt is marked for review

Scenario 3: Adding new authorized device
GIVEN an IT staff has access to the system
WHEN the account management screen is opened
THEN the IT staff can change the authorized device token for a user

USER INTERFACE

US-UIF-01: As a dashboard operator, I want to view a centralized, real-time interface displaying the status of all connected ADCs, active alerts, and relevant patient data, so that I can quickly assess the hospital's medication distribution health and respond to pending requests.

Scenario 1: Centralized dashboard monitoring
GIVEN the monitoring system is operational and communicating with multiple ADCs
WHEN an authorized dashboard operator logs into the system
THEN the system displays a high-level visual summary of all connected ADCs, active alarms categorized by severity, and pending tasks
AND all status indicators reflect changes within 2 seconds of an event occurring

Scenario 2: Navigating to specific ADC details
GIVEN the dashboard operator is viewing the main interface
WHEN the operator selects a specific ADC unit that is showing a "low inventory" warning
THEN the system navigates to a detailed view showing the exact medication slot, current count, and minimum threshold

Scenario 3: Handling UI timeouts for security
GIVEN the dashboard operator is logged into the interface
WHEN there has been no mouse or keyboard interaction for 2 minutes
THEN the system obscures the dashboard view and prompts the user to re-authenticate to restore access

US-UIF-02: As a nurse, I want to adjust my notification preferences through the user interface, so that I only receive alerts for my assigned ADC units and do not get overwhelmed by irrelevant alarms.

Scenario 1: Updating alert preferences
GIVEN a nurse is logged into the dashboard
WHEN they navigate to the notification settings page and select to receive only alerts for "ADC-01"
THEN the system updates their profile and stops routing alerts from other ADCs to their personal view

Scenario 2: Preventing unsafe configurations
GIVEN a user is modifying their notification settings
WHEN they attempt to disable all critical hardware fault alarms
THEN the user interface prevents the action and displays an error stating that critical safety alerts cannot be muted

Scenario 3: Session timeout during configuration
GIVEN a user is actively on the notification settings page
WHEN the user leaves the terminal and provides no input for 2 minutes
THEN the system automatically logs the user out and discards any unsaved notification preference changes to protect access

US-UIF-03: As a clinical supervisor, I want to manage and assign staff requests directly through my dashboard, so that I can ensure operational tasks and medication deliveries are handled efficiently.

Scenario 1: Manual task assignment
GIVEN a request is awaiting assignment from a clinical supervisor
WHEN the task is assigned to an available employee (pharmacist, nurse, IT worker)
THEN the task will appear on the employee’s dashboard

Scenario 2: Confusing details in request
GIVEN a request was sent AND the clinical supervisor is confused about its contents
WHEN the clinical supervisor returns the request
THEN the returned request will appear on the sender’s dashboard, specifying the source of confusion

Scenario 3: Routine task management
GIVEN the routine task list is inputted into the system
WHEN a task needs to be assigned
THEN the system will automatically assign it to the appropriate employee AND the task will appear on the employee's dashboard

US-UIF-04: As an IT administrator, I want to filter the system event logs displayed on my dashboard by specific parameters (like date, user, or event type), so that I can efficiently isolate and investigate system faults or security incidents.

Scenario 1: Filtering by a specific event type
GIVEN an IT administrator is viewing the event log interface
WHEN they apply a filter to only show "Failed Login Attempts"
THEN the UI dynamically updates to display only the events matching that specific category

Scenario 2: Viewing unreviewed security logs
GIVEN an IT administrator securely accesses their specific dashboard
WHEN the main dashboard page loads
THEN the system automatically displays a prioritized list of unreviewed log-in attempts

Scenario 3: Filter returns no results
GIVEN an IT administrator is actively searching the event log interface
WHEN they apply a filter for a specific date range or ADC unit where no events occurred
THEN the user interface displays a clear "No events found for this criteria" message instead of returning a system error or an unresponsive blank table

PATIENT INFORMATION DATABASE

US-PID-01: As a pharmacist, I want to securely query the patient information database for medication schedules and dispense histories, so that I can accurately track patient profiles and verify that the correct medications are being delivered.

Scenario 1: Retrieving a patient medication schedule
GIVEN the pharmacist is securely authenticated into the system
WHEN they query the database using a specific patient ID
THEN the system decrypts and retrieves the patient's active medication schedule and recent ADC dispensing history
AND returns the results to the user interface within 1 second

Scenario 2: Anonymizing data for external alerts
GIVEN a patient is scheduled for a dose that is missed by a nurse
WHEN the alarm manager requests data from the Patient Information Database to send a notification to personal devices
THEN the database provides the medication and location details but scrubs the patient's name and personal details from the payload

Scenario 3: Audit logging of database access
GIVEN a database query is executed for a patient profile
WHEN the data is successfully retrieved
THEN the database logs the user ID, timestamp, and specific records accessed for future accountability

US-PID-02: As a clinical supervisor, I want to view an audit log of who has accessed specific patient medication records in the database, so that I can ensure accountability and verify that patient privacy is being upheld.

Scenario 1: Generating an access audit
GIVEN a clinical supervisor has appropriate authorization
WHEN they request the access history for a specific patient ID
THEN the database retrieves a chronological log detailing the user ID, timestamp, and data accessed for that patient

Scenario 2: Unauthorized database query
GIVEN an unauthorized user or unauthenticated device attempts to query the database for a patient's medication details
WHEN the system receives the request
THEN the system denies access based on permissions and logs the failed attempt

Scenario 3: Querying a non-existent patient
GIVEN a pharmacist queries the database for a patient ID
WHEN the patient ID does not match any stored records (invalid input)
THEN the database rejects the request with a "Not Found" message without exposing or suggesting other patient names

US-PID-03: As a pharmacist, I want to retrieve historical dispensing data tied to specific patient profiles, so that I can audit previous medication activities and review historical operations.

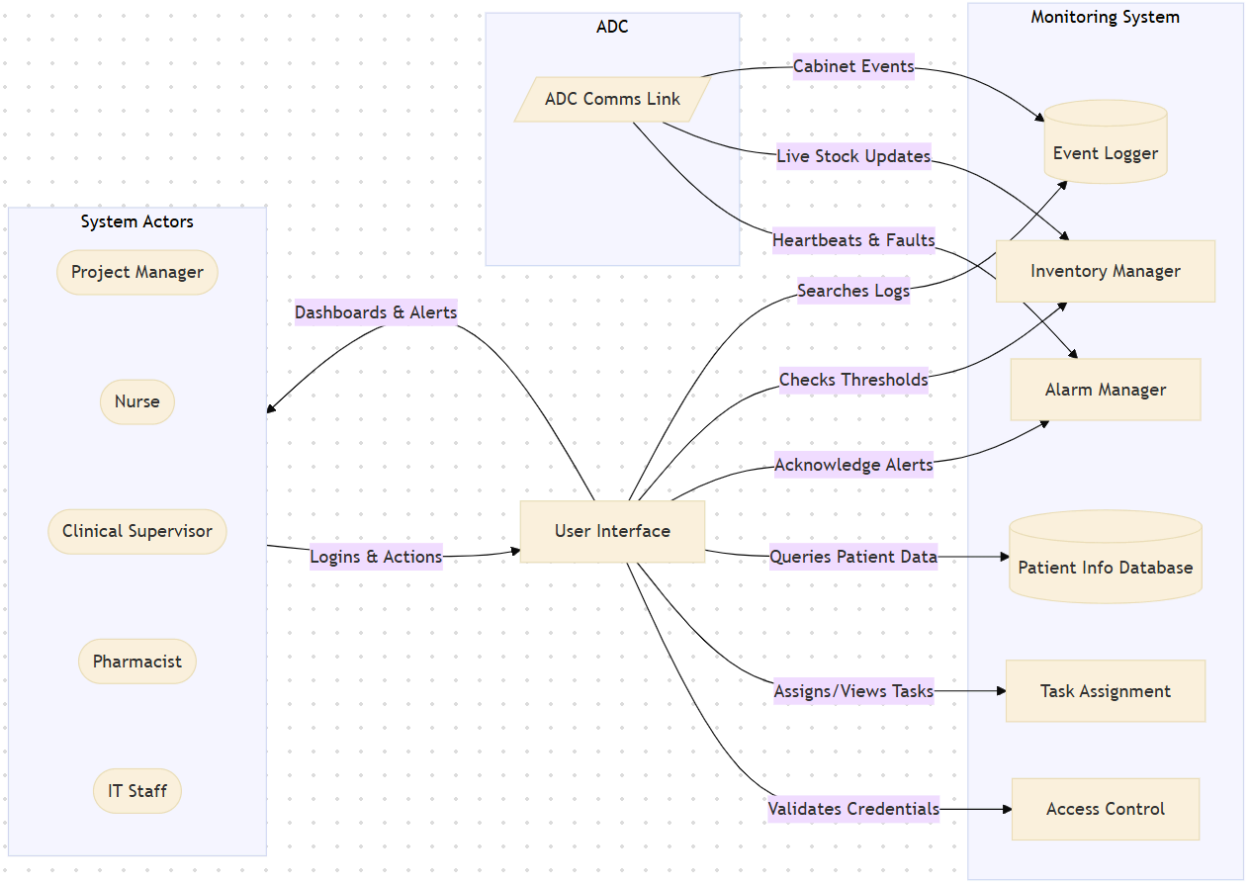
Scenario 1: Historical Review
GIVEN a pharmacist reviews previous dispensing activities from an ADC
WHEN they query the database for a specific patient's historical record
THEN the Monitoring System retrieves stored information and displays the requested data

Scenario 2: Exporting Historical Data
GIVEN the historical patient data is successfully retrieved and displayed on the screen
WHEN the user requests to export or print the report
THEN the database generates and provides a secure document containing the history

Scenario 3: Database query timeout
GIVEN the pharmacist queries a massive historical dataset spanning multiple years
WHEN the database query exceeds the maximum processing time due to data volume
THEN the system aborts the request, logs a timeout, and prompts the user to refine their search parameters

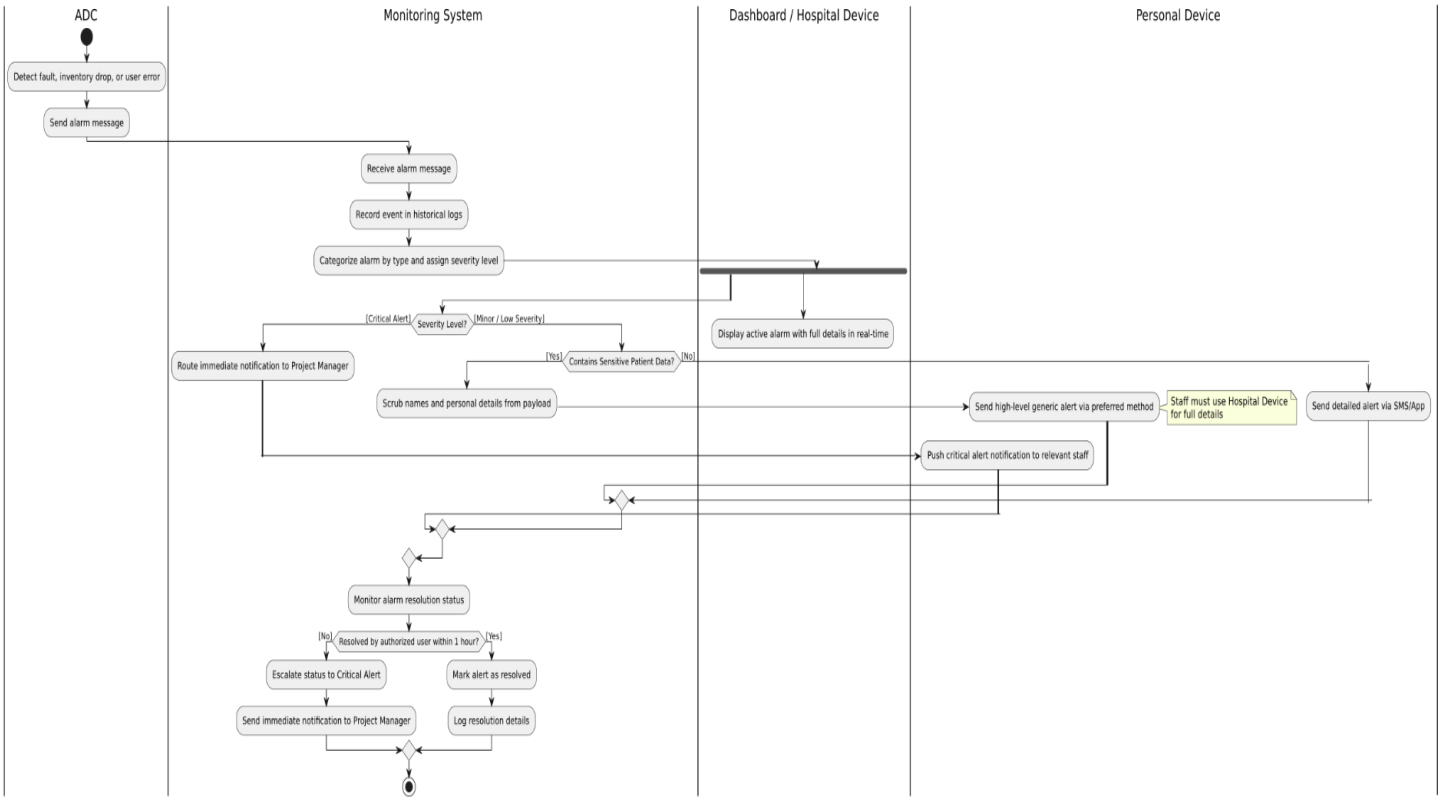
Diagrams

HIGH-LEVEL ARCHITECTURE DIAGRAM

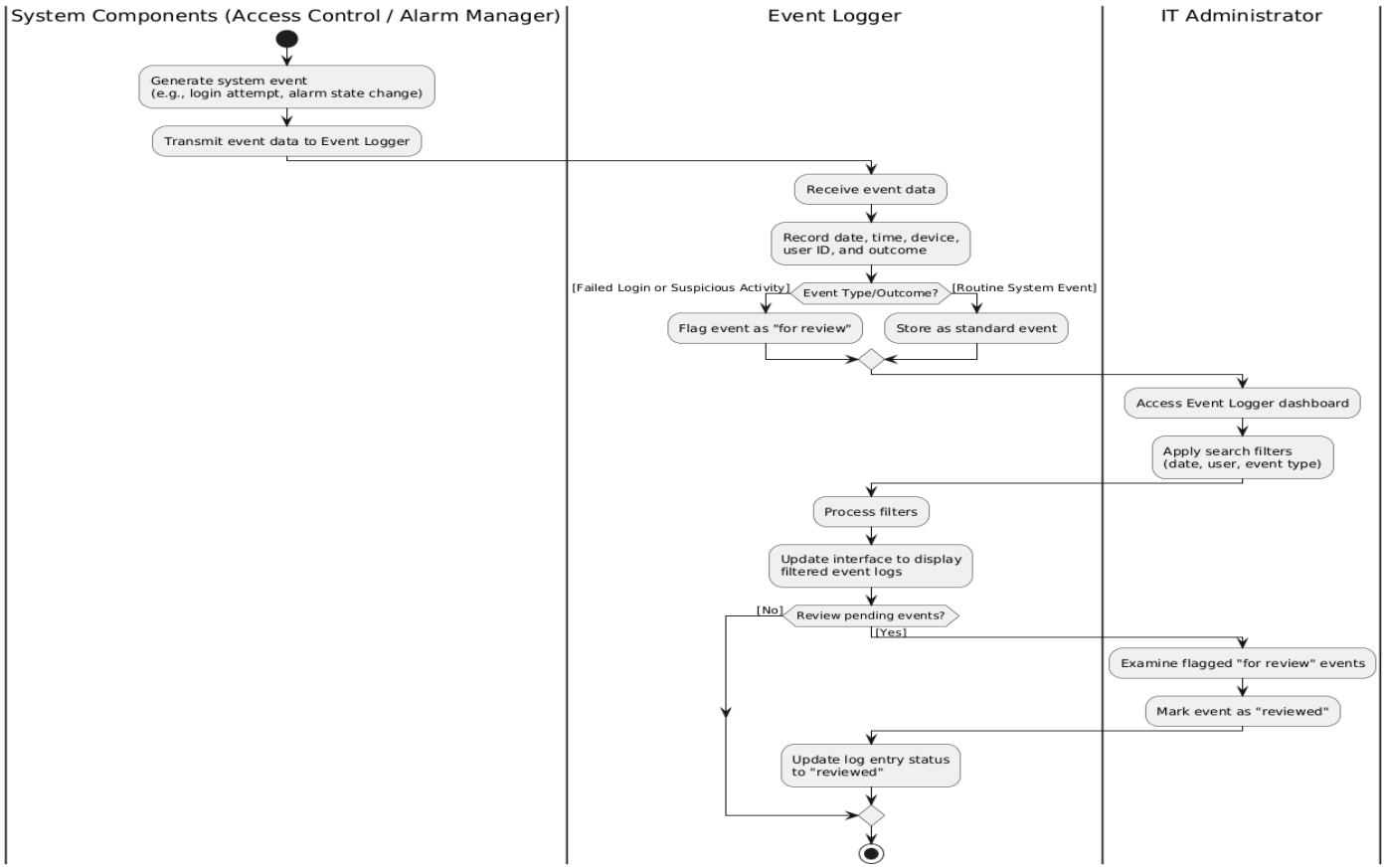


ACTIVITY DIAGRAMS

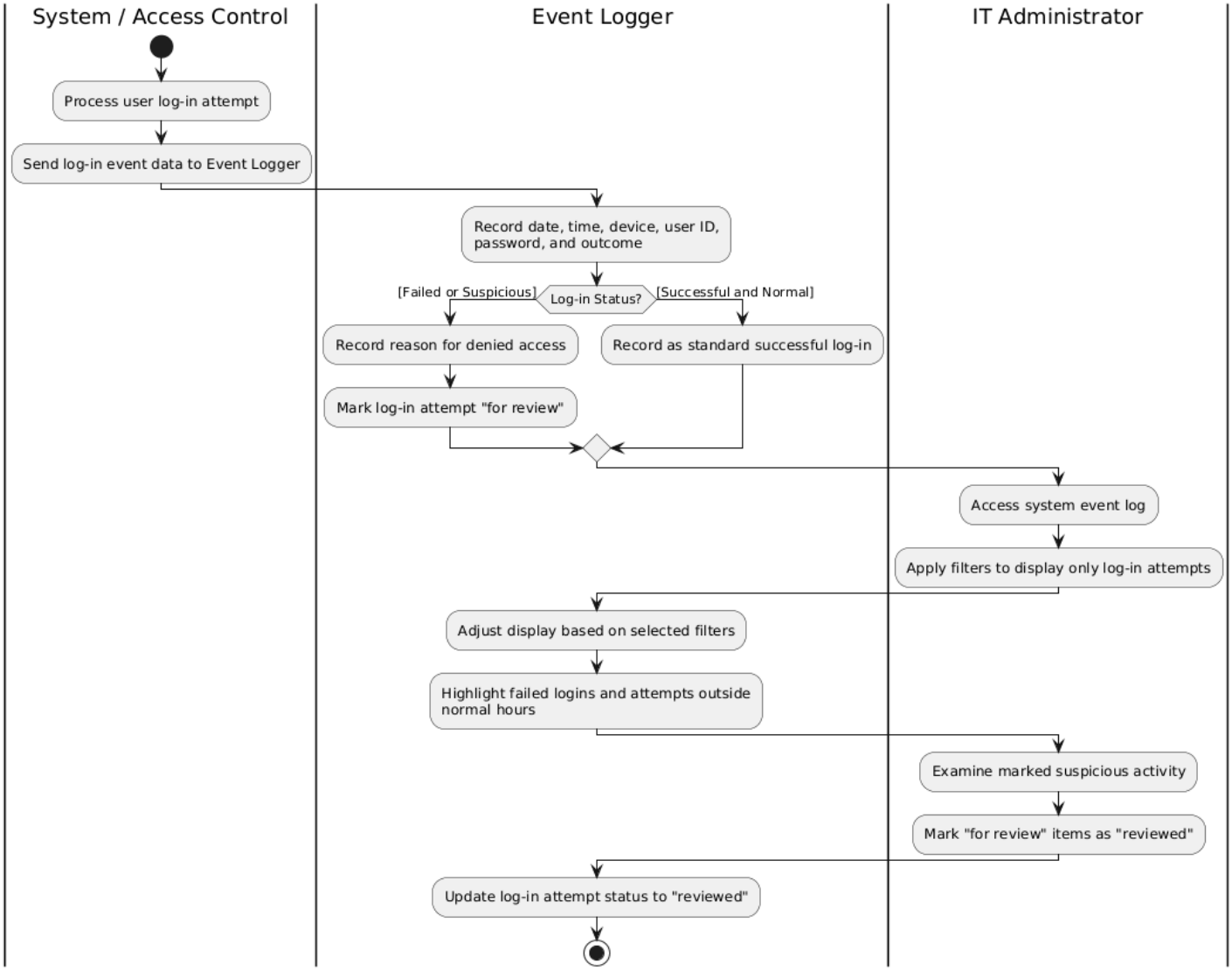
ALARM MANAGEMENT



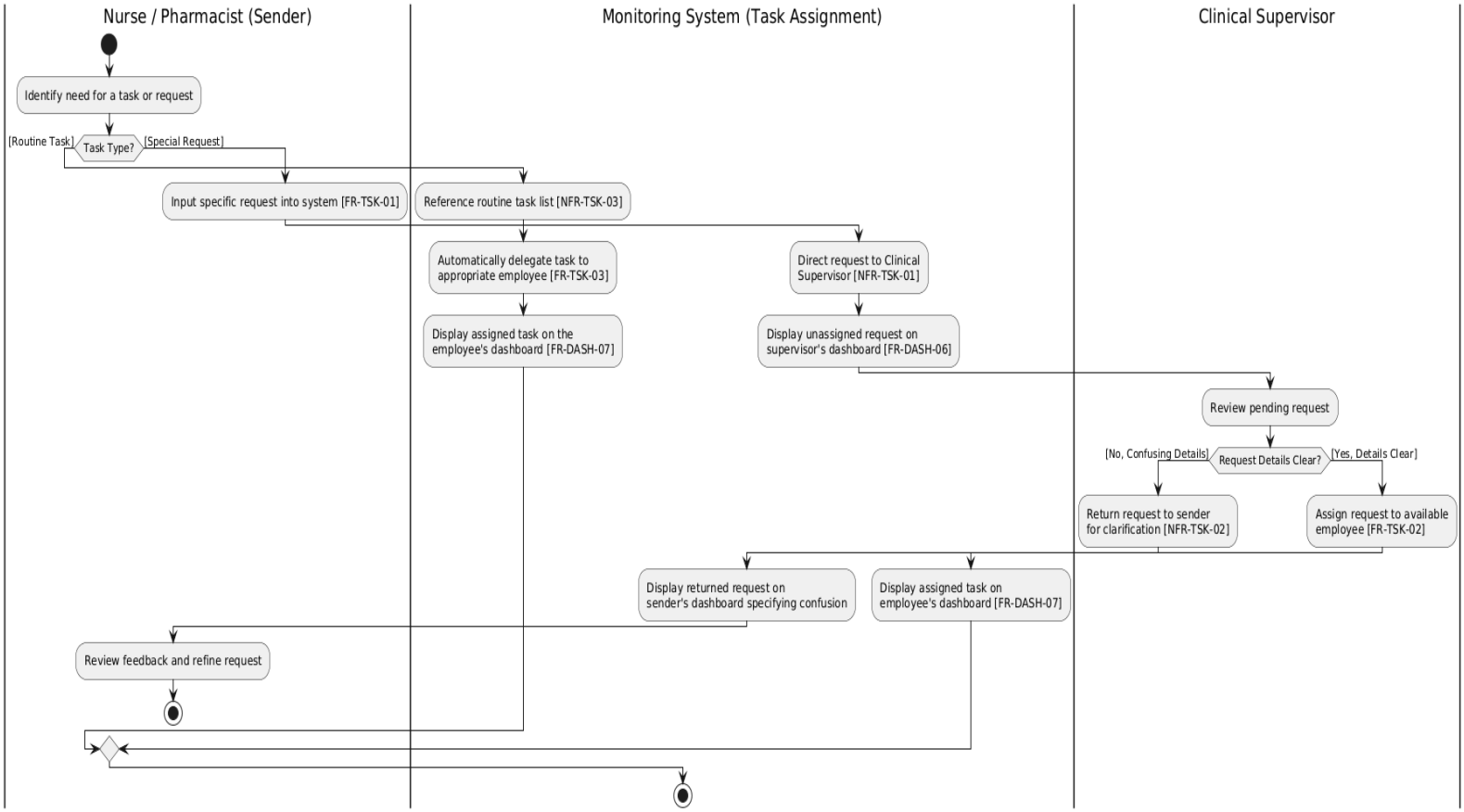
INVENTORY MANAGER



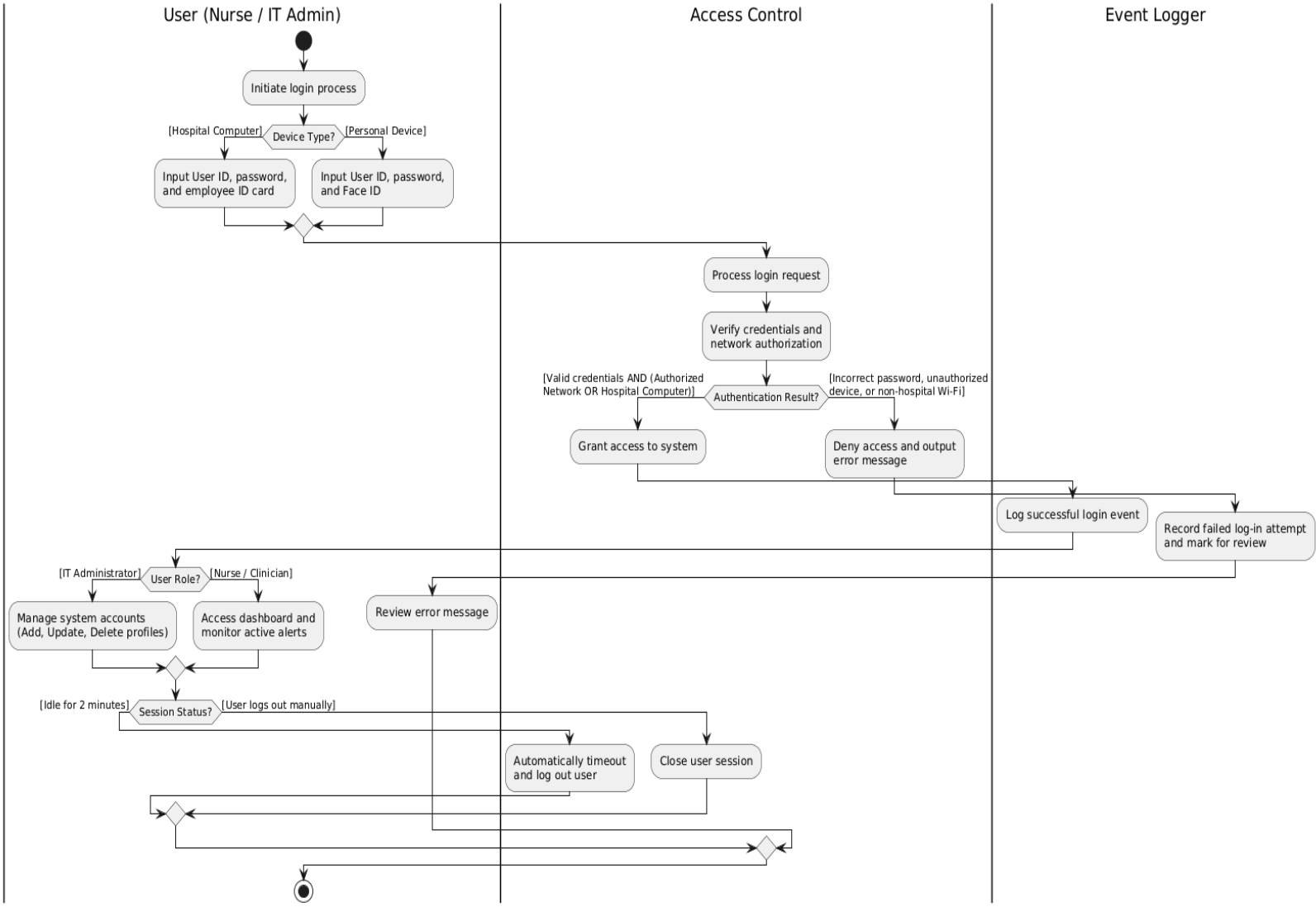
LOGGING


System / Access Control
Event Logger
IT Administrator
Process user log-in attempt
Send log-in event data to Event Logger
Record date, time, device, user ID, password, and outcome
Log-in Status?
[Failed or Suspicious]
[Successful and Normal]
Record reason for denied access
Record as standard successful log-in
Mark log-in attempt "for review"
Access system event log
Apply filters to display only log-in attempts
Adjust display based on selected filters
Highlight failed logins and attempts outside normal hours
Examine marked suspicious activity
Mark "for review" items as "reviewed"
Update log-in attempt status to "reviewed"

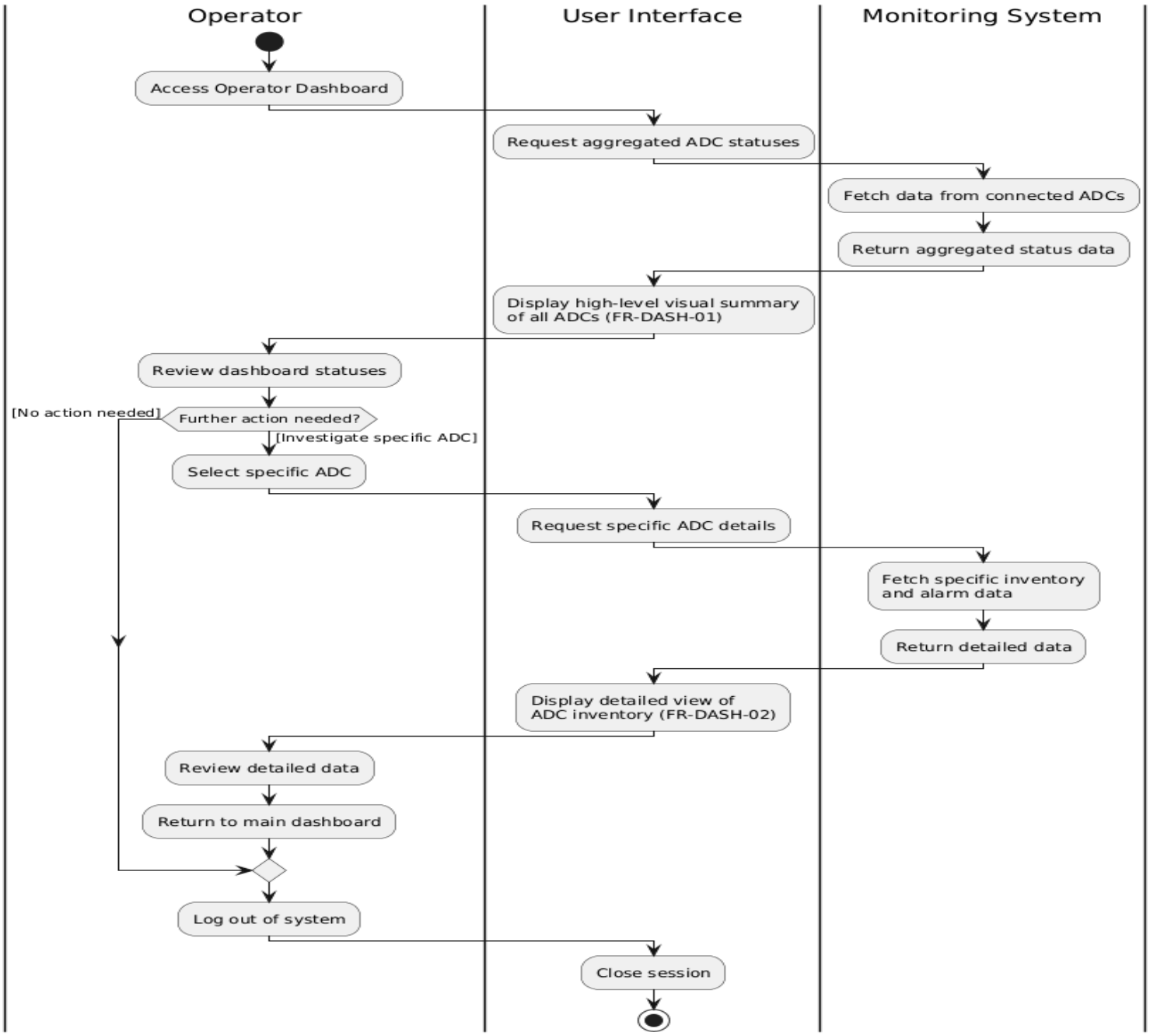
TASK ASSIGNMENT



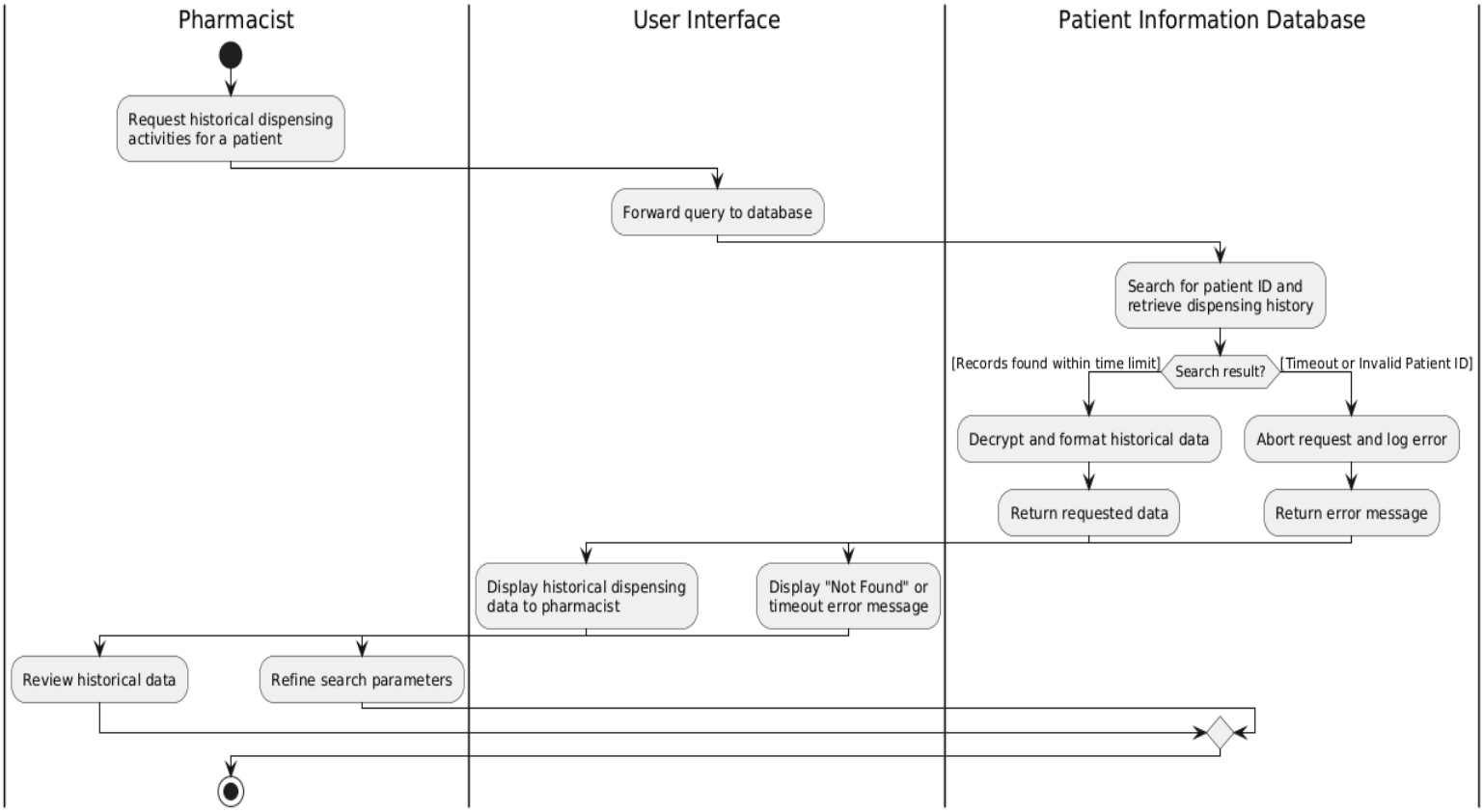
ACCESS CONTROL



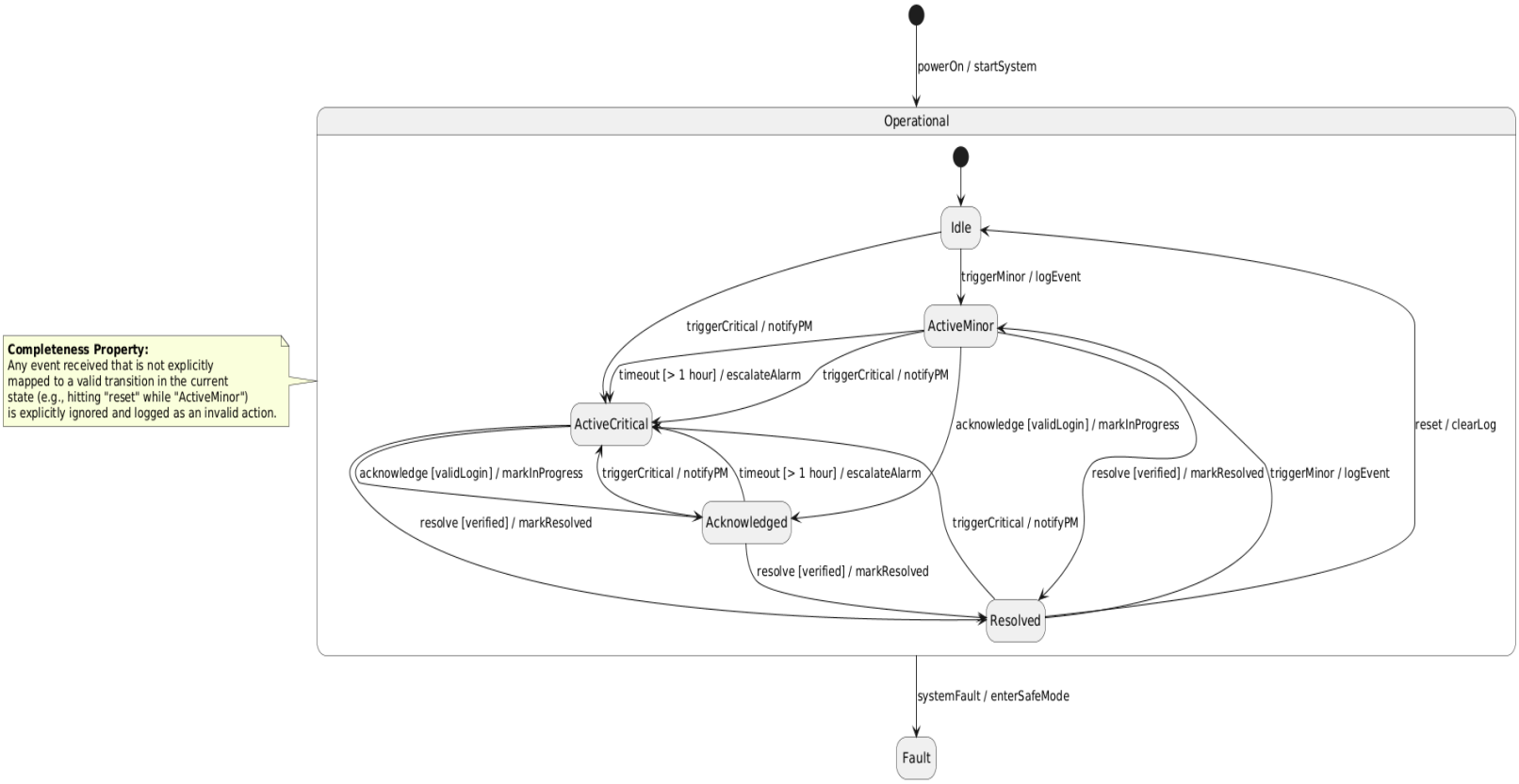
UI


Operator
User Interface
Monitoring System
Access Operator Dashboard
Request aggregated ADC statuses
Fetch data from connected ADCs
Return aggregated status data
Display high-level visual summary of all ADCs (FR-DASH-01)
Review dashboard statuses
[No action needed]
Further action needed?
[Investigate specific ADC]
Select specific ADC
Request specific ADC details
Fetch specific inventory and alarm data
Return detailed data
Display detailed view of ADC inventory (FR-DASH-02)
Review detailed data
Return to main dashboard
Log out of system
Close session

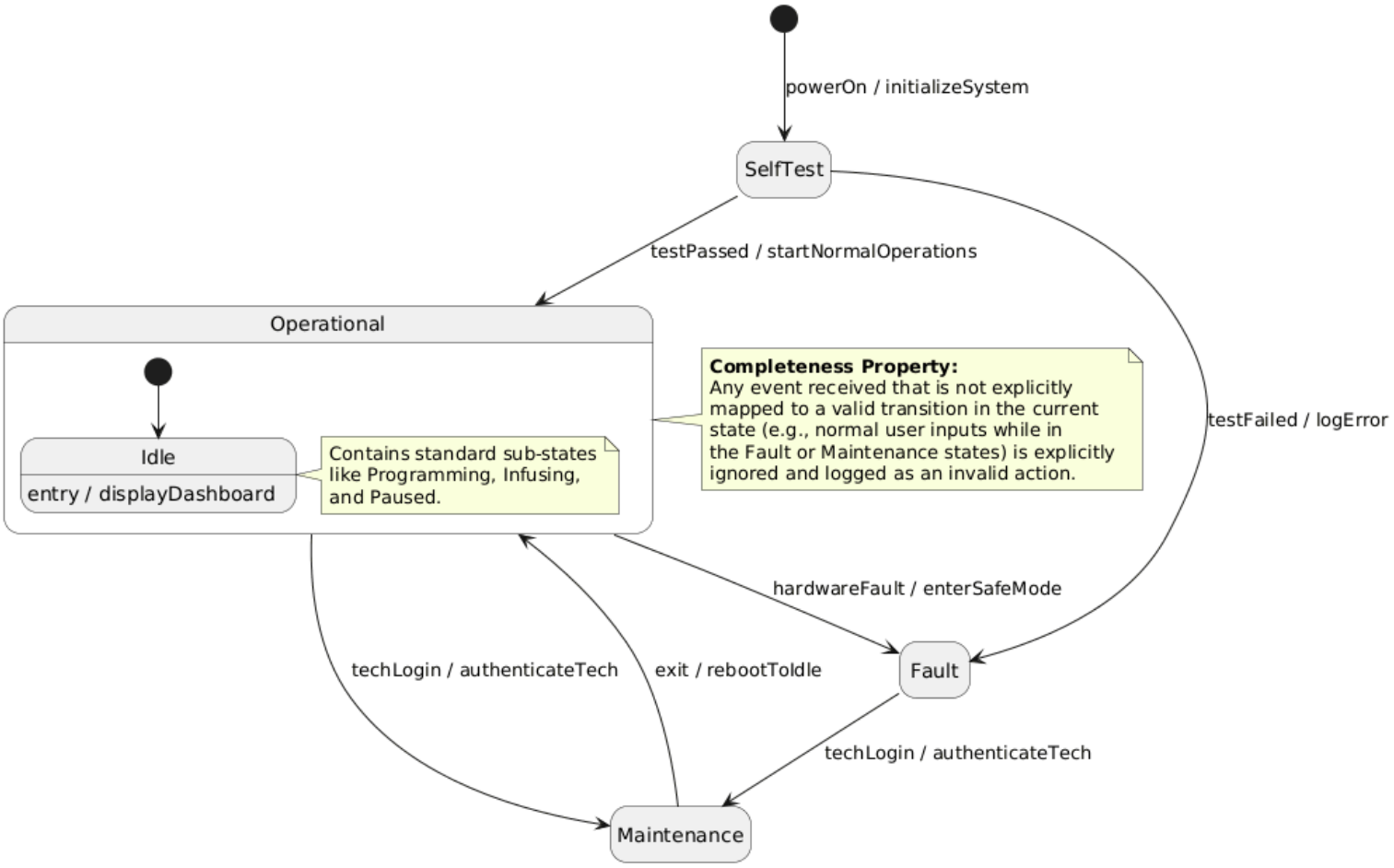
PATIENT INFORMATION DATABASE



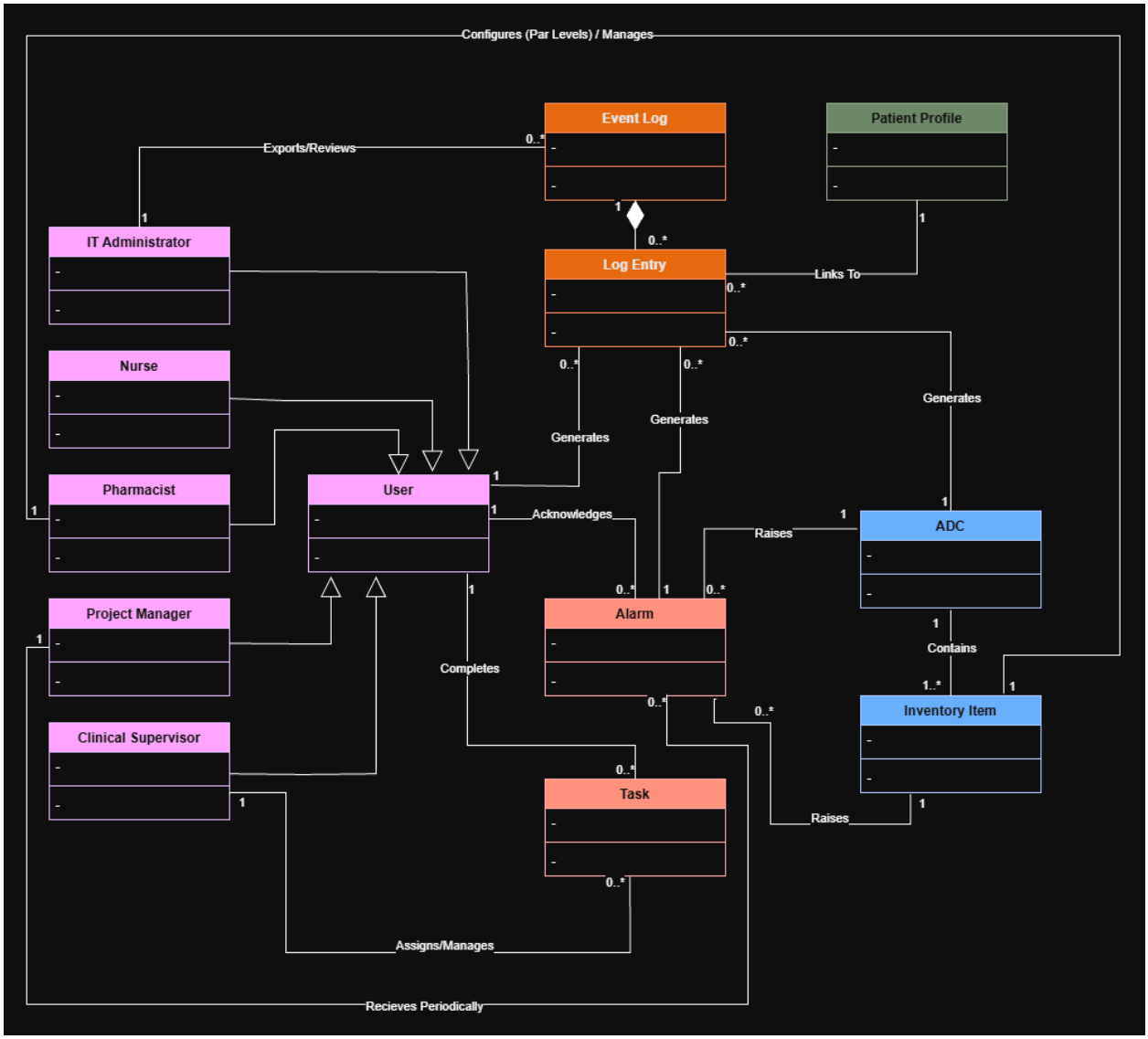
STATE DIAGRAM 1: ALARM LIFECYCLE



STATE DIAGRAM 2: SYSTEMS OPERATIONAL MODES



CONCEPTUAL DOMAIN MODEL



Traceability Table

ADC Communication				
Requirement ID	**Requirement (Short)**	**User Story ID(s)**	**Related Component**	**Stakeholder/Source**
FR-COM-01	Maintain connection	US-INV-01	ADC Comms Link	Pharmacist
FR-COM-02	Verify ADC connectivity	US-COM-01	ADC comms link	IT administrator
FR-COM-03	Alarm for failed connection	US-COM-01	Alarm Manager, ADC comms link	IT administrator
NFR-COM-01	Health check every 30 seconds	US-COM-01	ADC comms link	IT administrator
Alarm Management				
FR-ALM-01	Receive alarms/alerts from all ADC units	US-ALM-01, US-ALM-02	Alarm Manager, Comms Gateway	Project Manager, Nurse
FR-ALM-02	Direct alarms/alerts to appropriate individuals (visible through authorized login)	US-ALM-01, US-ALM-02	Alarm Manager, Dashboard Router	Project Manager, Nurse, IT Staff
FR-ALM-03	Send alerts to hospital staff via personal devices	US-ALM-02	Alarm Manager	Nurse
FR-ALM-04	Adjust alarm notification settings	US-ALM-02 US-UIF-02	Alarm Manager, User Interface	Nurse
NFR-ALM-01	Critical alert to project manager	US-ALM-01	Alarm Manager	Project Manager
NFR-ALM-02	Alert escalation after 1 hour	US-ALM-01	Alarm Manager	Project Manager
NFR-ALM-03	Active critical alert in place of 10 am update	US-ALM-01	Alarm Manager	Project Manager

NFR-ALM-04	Exclude patient information from alert notifications to personal devices	US-ALM-02	Alarm Manager	Nurse, Patient Privacy Standard
NFR-ALM-05	Alert for incomplete user profile	US-ADM-01	Alarm Manager, Access Control	IT Administrator
NFR-ALM-06	Retry unreceived alarm after 30 seconds on connection failure	US-ALM-02	Alarm Manager, Network Comms Link	Nurse, Clinical Supervisor
NFR-ALM-07	If alert fails to reach recipient it will be forwarded to clinical supervisor	US-ALM-02	Alarm Manager	Nurse, Clinical Supervisor
Inventory Management				
FR-INV-01	Aggregate ADC inventory	US-INV-01, US-INV-03	Inventory Manager	Pharmacist, IT Staff
FR-INV-02	Flag low inventory	US-INV-01, US-INV-03	Inventory Manager	Pharmacist, Clinical Supervisor
FR-INV-03	Generate inventory alarms	US-INV-01, US-INV-03	Inventory Manager, Alarm Manager	Pharmacist, Clinical Supervisor, Project Manager
FR-INV-04	Configure inventory par levels	US-INV-02	Inventory Manager	Pharmacist
FR-INV-05	Track inventory replenishment	US-INV-03	Inventory Manager	Pharmacist, IT Staff
FR-INV-06	Update cabinet status upon restock	US-INV-03	Inventory Manager	Pharmacist, IT Staff
NFR-INV-01	Inventory synchronization consistency	US-INV-01, US-INV-03	Inventory Manager	IT staff
NFR-INV-02	Restrict	US-INV-02	Inventory	IT Staff, Project

	modification of inventory records		Manager	Manager, Pharmacist
NFR-INV-03	Inventory processing performance	US-INV-01	Inventory Manager	IT Staff
NFR-INV-04	Non-critical alert for low inventory	US-INV-01, US-INV-03	Inventory Manager, Alarm Manager	Pharmacist, Clinical Supervisor, Project Manager
NFR-INV-05	Critical alert for low inventory	US-INV-01, US-INV-03	Inventory Manager, Alarm Manager	Pharmacist, Clinical Supervisor, Project Manager
Reporting				
FR-REP-01	Report to senior hospital staff	US-ALM-01	Alarm Manager	Project Manager
FR-REP-02	Download reports	US-PID-03	Patient Information Database, Event Logger	Pharmacist
NFR-REP-01	Daily update to project manager	US-ALM-01	Alarm Manager	Project Manager
NFR-REP-02	Daily update summarizes recent alerts	US-ALM-01	Alarm Manager	Project Manager
Dashboard				
Requirement ID	**Requirement (short)**	**User Story ID(s)**	**Related Component**	**Stakeholder/Source**
FR-DSH-02	Relevant ADC data on dashboard	US-INV-01	Inventory Manager	Pharmacist
FR-DSH-04	ADC status on dashboard	US-COM-01	User interface	IT administrator
FR-DSH-05	Log-in logs on IT administrator dashboard	US-ADM-02 US-UIF-04	Event logger, access control	IT administrator
FR-DSH-06	Requests on clinical supervisor	US-TSK-01 US-UIF-03	Task assignment	Clinical supervisor

	dashboard			
FR-DSH-07	Tasks on nurse dashboard	US-TSK-01	Task assignment	Clinical supervisor
FR-DSH-08	Tasks on pharmacist dashboard	US-TSK-01	Task assignment	Clinical supervisor
FR-DSH-09	Completed tasks on clinical supervisor dashboard	US-TSK-01	Task assignment	Clinical supervisor
NFR-DSH-01	Display low inventory alert within 5 seconds	US-INV-01	Alarm Manager, Inventory Manager	Pharmacist
NFR-DSH-02	Display connection loss within 5 seconds	US-COM-01, US-INV-01	Alarm Manager	IT administrator, Pharmacist
NFR-DSH-03	Display inventory updates within 5 seconds	US-INV-01, US-INV-03	Inventory Manager	IT staff
Logging				
FR-LOG-01	Record login attempts	US-ADM-02 US-PID-02	Event logger, access control	IT administrator
FR-LOG-02	Search capabilities for event logs	US-ADM-02, US-UIF-04, US-LOG-02	Event Logger	IT Staff, Pharmacist, Clinical Supervisor
FR-LOG-03	Log standard events	US-LOG-01	Event Logger	IT staff, Pharmacists, Clinical supervisor
FR-LOG-04	Log alarm resolutions/overrides	US-INV-01	Event Logger, Alarm Manager	IT Staff, Sam Delta Project Manager, Clinical supervisor
FR-LOG-05	Log maintenance activities	US-LOG-01	Event Logger	IT Staff
NFR-LOG-01	Record log-in details	US-ADM-02	Event logger, access control	IT administrator

NFR-LOG-02	Mark failed log-ins	US-ADM-02, US-ADM-03, US-ADM-04	Event logger, access control	IT administrator
NFR-LOG-03	Record reason for failed log-in	US-ADM-02	Event logger, access control	IT administrator
NFR-LOG-04	Mark suspicious log-ins	US-ADM-02	Event logger, access control	IT administrator
NFR-LOG-05	Update status of “reviewed” log-ins	US-ADM-02	Event logger	IT administrator
NFR-LOG-06	Immutable event logs	US-LOG-03	Event Logger	IT staff, Project Manager
NFR-LOG-07	Durable log storage	US-LOG-03	Event Logger	IT Staff
NFR-LOG-08	High-precision log timestamps	US-LOG-03	Event Logger, User Interface	IT Staff
NFR-LOG-09	Restrict log access	US-LOG-09	Task Assignment, Event logger	IT Staff, Project Manager
NFR-LOG-10	Retain inventory history	US-LOG-01	Inventory Manager, Event Logger	IT Administrator
Task Management				
FR-TSK-01	Submit requests	US-TSK-01	Task assignment	Clinical supervisor
FR-TSK-02	Respond to requests	US-TSK-01 US-UIF-03	Task assignment	Clinical supervisor
FR-TSK-03	Delegate routine tasks	US-TSK-01 US-UIF-03	Task assignment	Clinical supervisor
NFR-TSK-01	Requests to clinical supervisors	US-TSK-01	Task assignment	Clinical supervisor
NFR-TSK-02	Return request for clarification	US-TSK-01 US-UIF-03	Task assignment	Clinical supervisor
NFR-TSK-03	Reference routine task list	US-TSK-01	Task assignment	Clinical supervisor

NFR-TSK-04	Mark tasks complete	US-TSK-01	Task assignment	Clinical supervisor
Admin/IT				
FR-ADM-01	Create/delete login profiles	US-ADM-01	Access control	IT administrator
FR-ADM-02	Edit login profiles	US-ADM-01	Access control	IT administrator
FR-ADM-03	Log-in from computer	US-ADM-03	Access control	IT administrator
FR-ADM-04	Log-in from personal device	US-ADM-04	Access control	IT administrator
FR-ADM-05	Access timeout	US-ADM-03 US-UIF-02	Access control	IT administrator
FR-ADM-06	Deny access	US-ADM-03	Access control	IT administrator
NFR-ADM-01	Prompt user profile creation	US-ADM-01	Access control	IT administrator
NFR-ADM-02	Computer log-in credentials	US-ADM-03	Access control	IT administrator
NFR-ADM-03	“Credentials invalid” display	US-ADM-03	Access control	IT administrator
NFR-ADM-04	Personal device log-in credentials	US-ADM-04	Access control	IT administrator
NFR-ADM-05	Personal device log-in with device authorization	US-ADM-04	Access control	IT administrator
NFR-ADM-06	Personal device log-in with authorized network	US-ADM-04	Access control	IT administrator
NFR-ADM-07	Personal device authorization	US-ADM-04	Access control	IT administrator
User Interface				
FR-UIF-01	Provide a centralized operator dashboard	US-UIF-01	User Interface, Inventory Manager, Alarm Manager, Task Assignment	System Actors (Project Manager, Nurse, Clinical Supervisor, Pharmacist, IT

				Staff)
FR-UIF-02	Navigate to detailed inventories, schedules, and logs	US-UIF-01	User Interface, Inventory Manager, Patient Info Database, Event Logger	System Actors (Project Manager, Nurse, Clinical Supervisor, Pharmacist, IT Staff)
NFR-UIF-01	Display state changes within 2 seconds	US-UIF-01	User Interface, Alarm Manager, Inventory Manager	System Actors (Project Manager, Nurse, Clinical Supervisor, Pharmacist, IT Staff)
NFR-UIF-02	Obscure data and lock session after 2 minutes	US-UIF-01	User Interface, Access Control	IT Staff, Project Manager, (Patient Privacy Standard)
NFR-UIF-03	Prohibit disabling critical safety alarms	US-UIF-01 US-UIF-02	User Interface, Alarm Manager	Project Manager (Patient Safety)
NFR-UIF-04	Render filtered event logs within 3 seconds	US-UIF-04	User Interface, Event Logger	IT Staff, Project Manager
Patient Information Database				
FR-PID-01	Store patient data	US-PID-01	Patient Information Database, User Interface	Pharmacist
FR-PID-02	Scrub PII before external transmission	US-PID-01	Patient Information Database, Alarm Manager	Pharmacist, Project Manager, (Patient Privacy Standard)
FR-PID-03	Link historical	US-PID-03	Patient	Pharmacist

	logs to patient profiles		Information Database, Event Logger	
FR-PID-04	Fulfill queries in accordance with permissions	US-PID-02	Patient Information Database, Access Control	Clinical Supervisor, Project Manager
FR-PID-05	Display data	US-PID-03	Patient Information Database	Pharmacist
NFR-PID-01	Encrypt patient records at rest	US-PID-01	Patient Information Database	Pharmacist, Project Manager
NFR-PID-02	Support concurrent queries under 1 second	US-PID-01	Patient Information Database	Pharmacist
NFR-PID-03	Maintain tamper-proof audit trail of retrievals	US-PID-02	Patient Information Database, Event Logger	Clinical Supervisor, Project Manager
NFR-PID-04	Secure historical data against tampering	US-PID-03	Patient Information Database, Event Logger	Pharmacist

Assumptions & Decisions

1. Context & Scope Assumptions

ID	Category	Assumption / Gap Filled	Rationale & Impact
ASM-01	Network	Assume the hospital network infrastructure is available, but the ADC can operate locally and handle communication losses temporarily.	Shapes requirements for local event logging, reliability, and offline resilience during network drops.
ASM-02	Hardware	Assume physical cabinet hardware, electronic locks, drawers, sensors, and actuators already exist and provide software APIs.	Keeps mechanical design, drawer mechanisms, and internal hardware details strictly out of scope.
ASM-03	Users & Auth	Assume a means of capturing user credentials at the cabinet is available and users have valid accounts.	Enables the implementation of strict access control enforcement and role-based permissions for users.
ASM-04	Fault Handling	Assume at worst, issues are escalated to and guaranteed to be resolved by the project manager	Sets the extent to which issues/alerts are expected to be escalated by the system.
ASM-05	Data	Assume all medication data (dosage, storage requirements, administration instructions, etc) is available.	Allows use of medication to be automatically monitored

ASM-06	Staffing	Assume nurses and pharmacists are assigned specific ADC units for which to review alerts	Shapes the hierarchy for alarm management; allows better organization than all alarms being sent to all hospital employees
ASM-07	Network	Assume the monitoring system is capable of detecting when an alarm fails to be received by recipient device	Keeps software regarding detection of connection failure strictly out of scope.
ASM-08	Hardware	Assume all employees have a personal ID card that integrates with a tap system	Provides an option for multi-factor authentification
ASM-09	Security	Assume hospital security policies recommend a 2-minute timeout duration	Justifies the somewhat arbitrary choice of time frame
ASM-10	Data	Assume the monitoring system has access to shift schedules, notably can check (Y/N) if a specified employee is working	Provides a layer of security
ASM-11	Hardware	Assume that most employees possess a cell phone capable of a face ID check	Provides a layer of security

2. Architectural & Design Decisions

Decision ID	Topic	Choice Made	Alternatives Considered	Why We Chose This
DEC-LOG-001	Event Timestamp Precision	Dual-Precision Time: Track milliseconds behind the scenes, but only	1. Track Seconds Only: The central system might get confused or glitch	Prevents System Confusion: Tracking milliseconds behind the scenes ensures

		show normal seconds (YYYY-MM-DD HH:MM:SS) on screens and reports.	when many cabinets send updates at the exact same second. 2. Show Milliseconds to Everyone: Adds unnecessary numbers to screens, making reports messy and harder for staff to read.	the main system can sort rapid-fire events (like quick drawer taps or network re-connections) in the exact order they happened without crashing. Easier for Staff: Showing only seconds matches standard hospital clocks, security camera footage, and patient charts, making theft and error audits much simpler for pharmacy teams. No Extra Cost: Modern databases use the exact same amount of storage space whether we track milliseconds or not, so we get better system reliability for free.

3. Ethical & Safety Considerations

- Patient Safety: Prevent incorrect dispensing, unauthorized access, or hidden device failures because both the ADC and Monitoring System are safety-critical systems where malfunctions can cause direct harm.
- Data Privacy & Traceability: Ensure all operational activities, inventory adjustments, and system alarms are permanently recorded and restricted to authorized personnel (nurses, pharmacists, administrators) to maintain compliance and auditability.
- Controlled Access: Enforce strict authentication checks so that only permitted users can perform protected actions or access specific medications.

Accessibility: Multi-Channel Notification Flexibility (US-ALRM-02 / FR-ALRM-03): By giving staff the flexibility to receive alerts through multiple avenues (SMS, email, or designated app notifications), the system accommodates different user preferences, device types, and individual communication needs.

Cognitive Load Management (FR-ALRM-02): Directing alarms explicitly to the appropriate individuals based on roles prevents broadcast fatigue. In high-stress clinical environments, filtering out irrelevant alarms ensures that healthcare workers can focus on actionable alerts without cognitive overload.

Interface & Device Adaptability (FR-ALRM-03 and US-ALRM-02, Scenario 3): Allowing staff to manage and toggle their notification preferences through authorized devices ensures that the notification mechanism aligns with what each individual can effectively access and monitor while on shift.